WAN, VPN and NAT Troubleshooting

Practical Objective

Review a classmate's WAN, VPN, and NAT configuration, identify one misconfiguration using show and debug commands, and fix the issue.

Troubleshooting Procedure

1. Check Interface Status

Used to verify IP addresses and confirm that the WAN and LAN interfaces are up/up.

```
Router# show ip interface brief
```

2. Check Routing Table

Used to verify that the required networks and routes are present.

```
Router# show ip route
```

3. Check NAT

Used to verify whether NAT translations are being created and to inspect NAT operation.

```
Router# show ip nat translations
Router# show ip nat statistics
```

4. Check VPN Status

Used to verify the ISAKMP/IPsec security associations and VPN traffic.

```
Router# show crypto isakmp sa
Router# show crypto ipsec sa
```

5. Use Debug Commands

Debug output was observed while generating test traffic to identify where the connection was failing.

```
Router# debug ip nat
Router# debug crypto isakmp
Router# debug crypto ipsec
```

6. Stop Debugging

Used to safely stop all active debugging commands.

```
Router# undebug all
```

Misconfiguration Found

The LAN interface was not configured as an inside NAT interface. Because of this, NAT translations were not being created for LAN traffic.

Fix Applied

```
Router(config)# interface g0/0
Router(config-if)# ip nat inside
```

The WAN interface was verified as the NAT outside interface:

```
Router(config)# interface g0/1
Router(config-if)# ip nat outside
```

Verification

After applying the fix, a ping was generated from the LAN PC to the remote network. The NAT operation was then verified using:

```
Router# show ip nat translations
Router# show ip nat statistics
```